Millimeter Wave MIMO Channel Estimation using sub-6 GHz Out-of-Band Information

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***Abstract*—Next-generation wireless communication systems will incorporate millimeter wave (mmWave) as one of the key technologies to increase data rates. These forthcoming mmWave systems will integrate multiple-input multiple-output (MIMO) technology to ensure sufficient link margin and are expected to be deployed in conjunction with sub-6 GHz systems. Configuring a MIMO communication link usually relies on estimating channel state information (CSI), which is difficult to acquire at mmWave frequencies due to the low pre-beamforming signal-to-noise ratio (SNR). In this paper, we propose a novel approach to estimate mmWave MIMO channels by leveraging out-of-band information from a sub-6 GHz band. Utilizing this approach, we develop three channel estimation methods and compare these methods with one using only in-band information. We investigate the influence of the K-factor, the SNR and the number of antennas on the performance of the proposed channel estimation methods through simulations. Additionally, we compare these methods in terms of their computational complexity. Finally, we validate the performance of the proposed methods through channel measurements conducted in an outdoor environment. The results demonstrate that the proposed methods outperform in-band mmWave channel estimation in terms of spectral efficiency, particularly in scenarios of low SNR and high K-factor.**

***Index Terms*—millimeter-wave communication, MIMO, channel estimation, sub-6 GHz, out-of-band information**

I. Introduction

CONTEMPORARY wireless communication systems primarily operate in the sub-6 GHz bands. However, the sub-6 GHz frequency bands are struggling to meet the increasing demand for high data rates due to spectrum shortage. Fortunately, the mmWave bands (24 GHz – 300 GHz) [2] offer significantly more bandwidth, enabling high data rate transmissions. Therefore, mmWave communication has become part of the current generation of wireless systems [3], [4], with applications across various domains including cellular networks [5], [6], industrial setups [7], [8] and vehicle-to-everything (V2X) communications [9], [10].

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The benefits of employing mmWave systems are not obtained without drawbacks. Compared to the sub-6 GHz frequency band with its larger wavelength, the propagation characteristics for mmWave signals with the smaller wavelength are different [11]–[21]. For example, according to the Friis free-space equation [22], a mmWave signal at 25 GHz will encounter an isotropic path loss 20 dB higher than that of a signal at 2.5 GHz. To compensate for this frequency-dependent path loss and to ensure sufficient link margin, most mmWave systems will utilize large antenna arrays [23]. This opens up the opportunity to use MIMO directional beamforming [24]. However, the main challenge in utilizing antenna arrays is link establishment, which is achieved through either beam selection or channel estimation [25]. Link establishment is particularly challenging at mmWave frequencies, due to the low pre-beamforming SNR.

Moreover, mmWave systems are being deployed alongside sub-6 GHz systems [26] to enable multi-band communications and improve coverage [27]. Sub-6 GHz systems benefit from more favorable propagation conditions, resulting in higher pre-beamforming SNR. Hence, reliable sub-6 GHz out-of-band information can be leveraged to aid mmWave link establishment. This out-of-band information can be extracted without the need for additional dedicated measurements or pilot transmissions at the mmWave band, making it an efficient and cost-effective resource.

A. State of the Art

So far, several strategies to leverage sub-6 GHz out-of-band information as side information in the mmWave band have been proposed. Cheng *et al.* [28] introduce an iterative algorithm designed to jointly optimize transmit precoding vectors for both, mmWave and sub-6 GHz signals in full dual (scheduling over two frequency bands) and half dual (scheduling over one frequency band) transmission systems. In [29], Ohta *et al.* demonstrate the feasibility of mmWave link quality prediction based on 5 GHz band CSI fingerprinting at multiple locations. Burghal *et al.* [30] formulate the band assignment in dual-band systems as binary classification problems and propose supervised machine learning (ML) solutions based on recurrent neural networks (RNN). In [31], Chergui *et al.* introduce a semi-blind uplink/downlink decoupling concept for sub-6 GHz/mmWave fifth generation (5G) networks utilizing a support vector machine (SVM) algorithm. Moreover, Mismar *et al.* [32] propose a supervised ML algorithm to improve the success rate of handovers between the two frequencies using sub-6 GHz and mmWave prior channel measurements.

In addition, numerous approaches have been proposed to use sub-6 GHz out-of-band information to aid mmWave beam selection. Out-of-band measurements have been proposed and verified for steering mmWave beams in an indoor 60 GHz environment [33]. This approach involves extracting directional information from existing WiFi signals to minimize the overhead associated with beam steering at 60 GHz, confirming the benefit of utilizing out-of-band information. Ali *et al.* [34], [35] propose using spatial information extracted at sub-6 GHz to improve mmWave compressed beam selection. In another work, Ali *et al.* [36] use the sub-6 GHz channel covariance as an out-of-band side information for mmWave compressed covariance estimation. In [36], out-of-band covariance translation eliminates the need for in-band training entirely, whereas out-of-band aided covariance estimation relies on both, in-band and out-of-band training. Kyösti *et al.* [37] analyze the feasibility of using low-band channel information for coarse estimation of high-band beam directions. This comprehensive study demonstrates that leveraging spatial channel information from lower-frequency bands to aid mmWave beam steering is practical from a radio channel perspective. However, mmWave systems should not blindly rely on out-of-band information. Similarly, Hofer *et al.* [38] investigate the feasibility of utilizing sub-6 GHz propagation paths for mmWave link establishment by applying the moving virtual antenna array principle in single-antenna systems. Their findings reveal that, in line-of-sight (LOS) scenarios, link establishment with sufficient SNR in the mmWave band can be successfully achieved using angle-of-arrival (AoA) information from the sub-6 GHz band. Furthermore, Hashemi *et al.* [39] experimentally validate the correlation between the sub-6 GHz and mmWave AoA, particularly in the presence of the LOS component, in both indoor and outdoor environments. This study demonstrates that under LOS conditions, 94% of the measurements show the AoA identified in the sub-6 GHz band to be within $\pm 10^\circ$ of the AoA in the mmWave band. However, under non-line-of-sight (NLOS) conditions with a blocker present, the correlation between sub-6 GHz and mmWave signals decreases significantly and only 55% of the observations indicate that the strongest paths of the sub-6 GHz and mmWave signals align within $\pm 10^\circ$ of each other.

Besides conventional techniques, the literature contains a variety of mmWave beam selection approaches based on ML methods. In [40], Sim *et al.* introduce a beam selection algorithm that utilizes a sub-6 GHz power delay profile (PDP) to select an appropriate mmWave beam. Further, Ma *et al.* [41] predict the optimal mmWave beam by fusing sub-6 GHz channel information and mmWave low-overhead measurements. Moreover, Alrabeiah *et al.* [42] develop a deep learning model to predict the optimal mmWave beam and blockage status directly from the sub-6 GHz channel.

Analog and hybrid beamforming architectures use sequential beam scanning to search for the optimal beam direction. In analog systems, typically only one beam can be formed per training interval, while hybrid architectures allow a limited number of beams. This sequential process makes channel estimation time-consuming, introducing significant time overhead [43]. In contrast, fully digital beamforming architectures enable parallel beamforming, allowing channel estimation for multiple directions within a single training interval. As a result, fully digital architectures significantly reduce the time overhead for link establishment, making them more advantageous for latency-critical applications. To our best knowledge, there is no prior work that exploits sub-6 GHz out-of-band information to aid mmWave MIMO channel estimation with fully digital beamformers.

However, compared to analog or hybrid architectures, fully digital beamforming architectures for wideband mmWave systems can be significantly more power-hungry. This is primarily because each radio frequency (RF) chain requires a pair of analog-to-digital converters (ADC) at the receiver and digital-to-analog converters (DAC) at the transmitter. To keep the power consumption of these front ends low, fully digital beamformers employ converters with low resolution, typically 4 bits or fewer. Recent work [44] has demonstrated that ADC/DAC resolutions of 3 to 4 bits are sufficient to meet the transmitted signal requirements defined by the 3rd Generation Partnership Project (3GPP) specifications for wideband mmWave applications. Therefore, fully digital beamformers can be a feasible option for wideband mmWave systems, provided that the ADC/DAC resolutions remain sufficiently low.

B. Contributions

The main contributions of this paper are as follows:

- Extending our conference paper [1], we propose a novel approach to estimate mmWave CSI for MIMO beamforming wireless communication systems. The proposed approach relies on joint angle-of-departure (AoD) and AoA estimation for the LOS component using a Bartlett beamformer [45] in the sub-6 GHz band. The obtained AoD and AoA are then utilized to reconstruct the mmWave channel by exploiting the relationship between LOS channel components across different frequency bands. Unlike beam selection approaches, e.g. [43], our approach eliminates the need for time-consuming overhead in determining operational beam directions.
- Based on the proposed approach, we develop three methods for combining conventional in-band CSI and reconstructed out-of-band CSI to improve channel estimation accuracy for diverse link conditions. The translating method utilizes only out-of-band CSI, whereas the averaging method combines both in-band and out-of-band CSI, assigning equal weights of 0.5 to each. In contrast, the maximal ratio combining (MRC) method combines in-band and out-of-band CSI using an optimal weighting factor that dynamically adjusts based on the wireless channel conditions. Specifically, under LOS conditions, the MRC method relies more on out-of-band CSI,

leading to potential improvements in channel estimation accuracy. Conversely, under NLOS conditions (where the similarity between the sub-6 GHz and mmWave channels diminishes), the MRC method places more weight on in-band CSI, thereby maintaining the performance of the conventional mmWave system without degradation.

- We demonstrate the performance of the proposed methods by investigating the normalized channel estimation and interpolation mean squared error (MSE) and achievable spectral efficiency (SE) by means of Monte-Carlo simulations. Through these evaluations, we show the gains achieved compared to the channel estimation method using only in-band mmWave CSI. We investigate the influence of the K-factor, the SNR and the number of antennas on the performance of the proposed channel estimation methods. Additionally, we perform a comparative analysis of these methods based on their computational complexity and the associated time overhead. Furthermore, we validate the performance of the proposed methods through channel measurements obtained from a multi-band MIMO channel-sounding campaign conducted in an outdoor environment. Specifically, we utilize the measured sub-6 GHz and mmWave MIMO channels to evaluate the performance in terms of achievable SE as a function of SNR.

C. Organization

The rest of the paper is organized as follows. In Section II, we introduce the system and the channel model considered in this work. In Section III, we revisit the baseline in-band channel estimation methods. In Section IV, we provide a detailed description of the joint AoD and AoA estimation for the LOS component. In Section V, we describe the proposed channel estimation methods. The performance evaluation of the proposed methods based on simulations is presented in Section VI, while the evaluation based on measurements is detailed in Section VII. Finally, Section VIII provides concluding remarks for the paper.

D. Notation

The superscript $(\cdot)^{(\mathrm{b})}$ stands for frequency-band dependent values, where $\mathrm{b} \in \{\mathrm{s}, \mathrm{m}\}$. In this context, s stands for the sub-6 GHz frequency band and m stands for the mmWave frequency band. Further, we denote a scalar by x, a column vector by bold lowercase letters $\mathbf{x}$ and a matrix by bold uppercase letters $\mathbf{X}$. The i-th row of the matrix $\mathbf{X}$ is denoted by $\mathbf{X}_{i,:}$, and the j-th column of the matrix $\mathbf{X}$ is denoted by $\mathbf{X}_{:,j}$. The Kronecker product is denoted by $\otimes$ and the vectorization of a matrix is denoted by $\mathrm{vec}(\cdot)$. The superscript $(\cdot)^{\mathrm{T}}$ stands for transposition, the superscript $(\cdot)^{*}$ stands for complex conjugate and the superscript $(\cdot)^{\mathrm{H}}$ stands for Hermitian transposition. The notation $\|\cdot\|_F$ represents the Frobenius norm, $\|\cdot\|$ denotes the Euclidean norm and $\|\cdot\|_0$ represents the ℓ_0 norm. The delta function is denoted by $\delta[\cdot]$. The expectation of a random variable is denoted by $\mathbb{E}\{\cdot\}$.

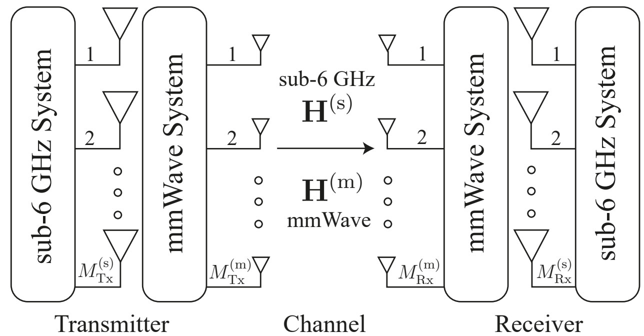

Fig. 1. Our multi-band MIMO transmission system with co-located sub-6 GHz and mmWave antenna arrays. $\mathbf{H}^{(\mathrm{s})}$ denotes the sub-6 GHz MIMO channel. $\mathbf{H}^{(\mathrm{m})}$ denotes the mmWave MIMO channel.

II. System Model

In this section, we introduce the wireless system model employed in this paper. We start by introducing our multi-band MIMO system. Subsequently, in Section II-A, we describe the Rician channel model adopted in this paper. Finally, in Section II-B, we explain the procedure of link establishment, which includes both, a training phase and data transmission.

We consider a point-to-point multi-band MIMO system, where sub-6 GHz and mmWave systems operate simultaneously in the radiative far-field regime. The transmitter consists of $M_{\mathrm{Tx}}^{(\mathrm{s})}$ sub-6 GHz and $M_{\mathrm{Tx}}^{(\mathrm{m})}$ mmWave antenna elements, while the receiver is equipped with an antenna array comprising $M_{\mathrm{Rx}}^{(\mathrm{s})}$ sub-6 GHz and $M_{\mathrm{Rx}}^{(\mathrm{m})}$ mmWave antenna elements (see Fig. 1). The sub-6 GHz and mmWave parts of the system are arranged as a uniform linear array (ULA) of dipole antennas, modeled as isotropic point sources. We assume that the sub-6 GHz and mmWave antenna arrays are co-located and aligned, resulting in identical AoD and AoA for both arrays in the LOS component [46] (see Fig. 2). In this setup, d denotes the distance between the center of the transmitter and the receiver, ϑ represents the AoD and φ represents the AoA. Both arrays have their antenna elements mutually separated by $\Delta d^{(\mathrm{b})} = 0.5\,\lambda^{(\mathrm{b})}$, where $\lambda^{(\mathrm{b})}$ represents the wavelength of the respective system. We consider propagation in the horizontal plane only, although generalization to 3D is possible by including a planar array instead, for example a uniform rectangular array. Furthermore, we assume perfect time and frequency synchronization between all transmitters and all receivers. The transmitter and receiver are equipped with one RF chain per antenna, thereby allowing for fully digital beamforming for both, sub-6 GHz and mmWave systems. While currently hybrid beamforming is used for mmWave systems, power-efficient fully digital beamformers have been proposed [44]. The transmission system operates using a time-division duplex (TDD) protocol, leading to reciprocal channel responses [47].

We consider an orthogonal frequency-division multiplexing (OFDM) system with $N^{(\mathrm{b})}$ subcarriers. Blocks of $N^{(\mathrm{b})}$ symbols modulated by quadrature amplitude modulation (QAM) are mapped onto $N^{(\mathrm{b})}$ different subcarriers to construct OFDM symbols. The OFDM system converts a broadband frequency-selective channel into narrowband frequency flat channels with the help of a discrete Fourier transform (DFT) and application

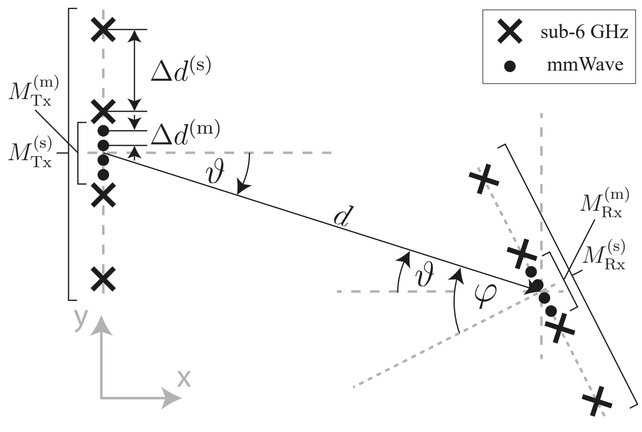

Fig. 2. The geometry of the transmit and receive ULA elements is characterized by their mutually separation $\Delta d^{(\mathrm{b})}$ along with corresponding AoD ϑ and AoA φ.

of a cyclic prefix [48].

A. Channel Model

The channel at the OFDM subcarrier n is described by an $M_{\mathrm{Rx}}^{(\mathrm{b})} \times M_{\mathrm{Tx}}^{(\mathrm{b})}$ dimensional complex-valued channel matrix $\mathbf{H}^{(\mathrm{b})}[n]$, employing the equivalent complex baseband representation of the OFDM system. We consider a frequency-selective Rician fading channel modeled as [22]

$$\begin{aligned}\mathbf{H}^{(\mathrm{b})}[n] =& \sqrt{\eta^{(\mathrm{b})}}\sqrt{\frac{K^{(\mathrm{b})}}{1+K^{(\mathrm{b})}}}\underbrace{\mathbf{H}_{\mathrm{fs}}^{(\mathrm{b})}}_{\text{free-space}} \\ &+\sqrt{\eta^{(\mathrm{b})}}\sqrt{\frac{1}{1+K^{(\mathrm{b})}}}\underbrace{\mathbf{H}_{\mathrm{rp}}^{(\mathrm{b})}[n]}_{\text{Rayleigh part}},\end{aligned} \tag{1}$$

where $\eta^{(\mathrm{b})}$ is the path gain including shadowing and $K^{(\mathrm{b})}$ denotes the Rician K-factor in the frequency band $\mathrm{b} \in \{\mathrm{s}, \mathrm{m}\}$. Considering the distinct characteristics of multipath components between mmWave and sub-6 GHz bands, we assume that the mmWave K-factor $K^{(\mathrm{m})} = c_K K^{(\mathrm{s})}$, where c_K denotes the scaling factor.

The deterministic free-space LOS channel is defined by

$$\mathbf{H}_{\mathrm{fs}}^{(\mathrm{b})} = e^{j\chi^{(\mathrm{b})}}\mathbf{a}_{\mathrm{Rx}}^{(\mathrm{b})}(\varphi)\left(\mathbf{a}_{\mathrm{Tx}}^{(\mathrm{b})}(\vartheta)\right)^{\mathrm{H}}, \tag{2}$$

where

$$\mathbf{a}_{\mathrm{Tx}}^{(\mathrm{b})}(\vartheta) = \begin{bmatrix}1 & \dots & e^{-j2\pi\left(M_{\mathrm{Tx}}^{(\mathrm{b})}-1\right)\frac{\Delta d^{(\mathrm{b})}}{\lambda^{(\mathrm{b})}}\sin(\vartheta)}\end{bmatrix}^{\mathrm{T}} \tag{3}$$

is the steering vector for the AoD ϑ and

$$\mathbf{a}_{\mathrm{Rx}}^{(\mathrm{b})}(\varphi) = \begin{bmatrix}1 & \dots & e^{-j2\pi\left(M_{\mathrm{Rx}}^{(\mathrm{b})}-1\right)\frac{\Delta d^{(\mathrm{b})}}{\lambda^{(\mathrm{b})}}\sin(\varphi)}\end{bmatrix}^{\mathrm{T}} \tag{4}$$

is the steering vector for the AoA φ. In Eq. (2), $\chi^{(\mathrm{b})} \sim \mathcal{U}(-\pi, \pi)$ denotes the random phase shift at the first (reference) antenna element.

The Rayleigh channel matrix $\mathbf{H}_{\mathrm{rp}}^{(\mathrm{b})}[n]$ is modeled according to the 3GPP channel model [49]. Given the large difference between the sub-6 GHz and mmWave wavelengths, we consider the Rayleigh channels $\mathbf{H}_{\mathrm{rp}}^{(\mathrm{s})}[n]$ and $\mathbf{H}_{\mathrm{rp}}^{(\mathrm{m})}[n]$ to be completely decorrelated. The delay-domain channel matrix $\mathbf{H}^{(\mathrm{b})}[\tau]$ for delay tap τ is then obtained by applying the inverse discrete Fourier transform (IDFT) to $\mathbf{H}^{(\mathrm{b})}[n]$.

The path loss for mmWave is defined by

$$\eta^{(\mathrm{m})} = \left(\frac{4\pi d f_{\mathrm{c}}^{(\mathrm{m})}}{c_0}\right)^2 = \left(\frac{4\pi d\sqrt{\alpha} f_{\mathrm{c}}^{(\mathrm{s})}}{c_0}\right)^2 = \alpha\eta^{(\mathrm{s})}, \tag{5}$$

where c_0 is the speed of light, $\eta^{(\mathrm{s})}$ represents the free space path loss for the sub-6 GHz band and $\sqrt{\alpha} = f_{\mathrm{c}}^{(\mathrm{m})}/f_{\mathrm{c}}^{(\mathrm{s})}$ denotes the carrier frequency ratio between the sub-6 GHz band and the mmWave band. We assume a pathloss exponent of 2, though it can be modified to account for different propagation environments. Since mmWave systems tend to use larger bandwidth and can have a noise figure different to sub-6 GHz systems, we introduce

$$\beta = \frac{B^{(\mathrm{m})}F^{(\mathrm{m})}P_{\mathrm{T}}^{(\mathrm{s})}}{B^{(\mathrm{s})}F^{(\mathrm{s})}P_{\mathrm{T}}^{(\mathrm{m})}} \tag{6}$$

as the ratio of sub-6 GHz and mmWave bandwidths B, noise figures F and transmit powers P_{T}. Then, the obtained pre-beamforming mmWave SNR for the single receive antenna element is expressed by

$$\begin{aligned}\gamma^{(\mathrm{m})} =& \frac{P_{\mathrm{T}}^{(\mathrm{m})}}{\eta^{(\mathrm{m})}k_{\mathrm{B}}T_{\mathrm{Rx}}B^{(\mathrm{m})}F^{(\mathrm{m})}} = \frac{P_{\mathrm{T}}^{(\mathrm{s})}}{\alpha\eta^{(\mathrm{s})}k_{\mathrm{B}}T_{\mathrm{Rx}}\beta B^{(\mathrm{s})}F^{(\mathrm{s})}} \\ =& \gamma^{(\mathrm{s})}\frac{1}{\alpha\beta},\end{aligned} \tag{7}$$

where k_{B} represents the Boltzmann constant and T_{Rx} denotes the antenna temperature. A sub-6 GHz system with $\alpha\beta$ times higher SNR has benefits. These benefits will be highlighted below.

B. Link Establishment

Establishing a communication link consists of a training phase and a data transmission phase. The wireless channel is estimated at sub-6 GHz and mmWave frequency bands during the training phase. The channel estimates obtained at the sub-6 GHz and the mmWave band are then utilized jointly for data transmission at the mmWave band.

1) Training Phase: The training phase comprises two consecutive steps. In the first step, non-precoded pilot symbols are transmitted in both frequency bands to acquire initial channel estimates. The second step then employs precoded and combined pilot symbols in the mmWave band only, leveraging the information obtained from the initial channel estimates.

During the first step of the training phase, each transmit antenna is assigned pilot symbols that are known at the receiver. The pilot symbols $\boldsymbol{\phi}^{(\mathrm{b})}[n] \in \mathbb{C}^{M_{\mathrm{Tx}}^{(\mathrm{b})}\times 1}$ generated from the QAM alphabet are distributed over $N^{(\mathrm{b})}$ subcarriers so that each antenna occupies a certain number of non-overlapping subcarriers. The pilot allocation at the t-th transmit antenna is given by

$$\phi_t^{(\mathrm{b})}[n] = \begin{cases}\phi^{(\mathrm{b})}[n], \; n \in \{t, t+M_{\mathrm{Tx}}^{(\mathrm{b})}, \dots, N^{(\mathrm{b})} - M_{\mathrm{Tx}}^{(\mathrm{b})} + t\} \\ 0, \quad \text{else}\end{cases} \tag{8}$$

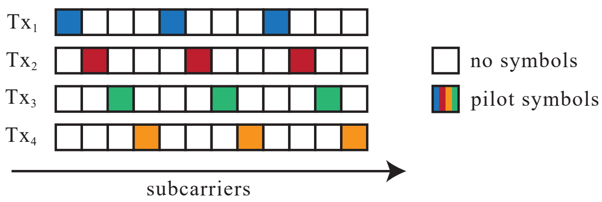

Fig. 3. Pilot symbols at each antenna occupy a certain number of subcarriers such that they do not overlap in the frequency domain.

where $t \in \{1, \ldots, M_{\mathrm{Tx}}^{(\mathrm{b})}\}$ denotes the transmit antenna index. A pilot allocation example for $M_{\mathrm{Tx}}^{(\mathrm{b})} = 4$ transmit antennas is shown in Fig. 3. The input-output relationship for the first step of the training phase is given by

$$\mathbf{y}^{(\mathrm{b})}[n] = \mathbf{H}^{(\mathrm{b})}[n]\boldsymbol{\phi}^{(\mathrm{b})}[n] + \mathbf{w}^{(\mathrm{b})}[n], \tag{9}$$

where the received signal is denoted by $\mathbf{y}^{(\mathrm{b})}[n] \in \mathbb{C}^{M_{\mathrm{Rx}}^{(\mathrm{b})} \times 1}$ and the additive white Gaussian noise (AWGN) with the power of $\sigma^2_{w^{(\mathrm{b})}}$ is denoted by $\mathbf{w}^{(\mathrm{b})}[n] \in \mathcal{CN}(0, \sigma^2_{w^{(\mathrm{b})}} \mathbf{I}_{M_{\mathrm{Rx}}^{(\mathrm{b})}})$. At the receiver, the least-squares (LS) channel estimates are obtained at the pilot positions $n_t \in \{t, t + M_{\mathrm{Tx}}^{(\mathrm{b})}, \ldots, N^{(\mathrm{b})} - M_{\mathrm{Tx}}^{(\mathrm{b})} + t\}$ defined by

$$\widetilde{\phi}_{r,t}^{(\mathrm{b})}[n_t] = \frac{\mathrm{y}_r^{(\mathrm{b})}[n_t]}{\phi_t^{(\mathrm{b})}[n_t]}, \tag{10}$$

where $r \in \{1, \ldots, M_{\mathrm{Rx}}^{(\mathrm{b})}\}$ represents the receive antenna index. Subsequently, linear interpolation is applied to estimate the channel at non-occupied subcarrier positions. As a result, we obtain the estimated channel $\widetilde{\mathbf{H}}^{(\mathrm{b})}[n] \in \mathbb{C}^{M_{\mathrm{Rx}}^{(\mathrm{b})} \times M_{\mathrm{Tx}}^{(\mathrm{b})}}$, with values for r-th receive and t-th transmit antenna given by

$$\widetilde{\mathrm{H}}_{r,t}^{(\mathrm{b})}[n] = \widetilde{\phi}_{r,t}^{(\mathrm{b})}[n_t] + (n - n_t)\, \frac{\widetilde{\phi}_{r,t}^{(\mathrm{b})}[n_t + M_{\mathrm{Tx}}^{(\mathrm{b})}] - \widetilde{\phi}_{r,t}^{(\mathrm{b})}[n_t]}{M_{\mathrm{Tx}}^{(\mathrm{b})}}. \tag{11}$$

Note that the channel estimation quality can be enhanced by employing linear minimum mean square error (LMMSE) estimator [50] and more sophisticated interpolation techniques [51], though this comes at the cost of higher computational complexity. As the channel is estimated at the receiver, the CSI has to be known also at the transmitter for optimal beamforming matrix selection. Since the system operates using the TDD protocol, CSI acquisition at the transmitter is achieved by exploiting channel reciprocity in both frequency bands.

During the second step of the training phase, the estimated sub-6 GHz channel $\widetilde{\mathbf{H}}^{(\mathrm{s})}[n]$ is firstly used to estimate the AoD $\widetilde{\vartheta}$ and AoA $\widetilde{\varphi}$, employing the method detailed in Section IV. This enables the pilot symbols to be precoded and combined in the direction of the LOS component. The input-output relationship for the second step of the training phase is then given by

$$\begin{aligned}\mathrm{y}^{(\mathrm{m})}[n] = &\left(\mathbf{a}_{\mathrm{Rx}}^{(\mathrm{m})}\left(\widetilde{\varphi}\right)\right)^{\mathrm{H}} \mathbf{H}^{(\mathrm{m})}[n]\, \mathbf{a}_{\mathrm{Tx}}^{(\mathrm{m})}\left(\widetilde{\vartheta}\right) \phi^{(\mathrm{m})}[n] \\ &+ \left(\mathbf{a}_{\mathrm{Rx}}^{(\mathrm{m})}\left(\widetilde{\varphi}\right)\right)^{\mathrm{H}} \mathbf{w}^{(\mathrm{m})}[n],\end{aligned} \tag{12}$$

where $\phi^{(\mathrm{m})}[n]$ and $\mathrm{y}^{(\mathrm{m})}[n]$ denote one-dimensional complex-valued quantities, as only a single spatial stream is employed. Following this, the receiver performs LS estimation and linear interpolation to obtain the effective beamformed scalar channel estimate $\widehat{\mathrm{G}}^{(\mathrm{m})}[n]$. To estimate the LOS component of $\widehat{\mathrm{G}}^{(\mathrm{m})}[n]$, we focus exclusively on the zero-delay channel tap, suppressing all other taps as they do not carry LOS information. First, we obtain the delay-domain channel estimate $\widehat{\mathrm{G}}^{(\mathrm{m})}[\tau]$ by applying the IDFT to $\widehat{\mathrm{G}}^{(\mathrm{m})}[n]$. Next, we filter this estimate to isolate the LOS component:

$$\widehat{\mathrm{G}}_{\mathrm{LOS}}^{(\mathrm{m})}[\tau] = \widehat{\mathrm{G}}^{(\mathrm{m})}[\tau]\delta[\tau] \tag{13}$$

At this point, we assume that the system is already time synchronized, such that the LOS component appears at $\tau = 0$. It is important to note that this isolation may not be perfect and could include some residual unfiltered channel components. Finally, we convert the LOS channel estimate back to the frequency domain by applying the DFT to $\widehat{\mathrm{G}}_{\mathrm{LOS}}^{(\mathrm{m})}[\tau]$. The resulting out-of-band aided mmWave MIMO channel estimate for the LOS component is then reconstructed as

$$\widehat{\mathbf{H}}^{(\mathrm{m})}[n] = \mathbf{a}_{\mathrm{Rx}}^{(\mathrm{m})}\left(\widetilde{\varphi}\right) \widehat{\mathrm{G}}_{\mathrm{LOS}}^{(\mathrm{m})}[n] \left(\mathbf{a}_{\mathrm{Tx}}^{(\mathrm{m})}\left(\widetilde{\vartheta}\right)\right)^{\mathrm{H}}. \tag{14}$$

2) Data Transmission: For the data transmission phase, we consider only the mmWave system. The estimated channels quantities from the training phase are processed using three methods, which will be proposed in Section V. The resulting mmWave channel estimate $\overline{\mathbf{H}}^{(\mathrm{m})}[n]$ is then utilized for calculating the precoding and combining. In this work, we consider singular value decomposition (SVD) for achieving the highest performance of the MIMO channel in terms of SE. The compact-form SVD of the channel matrix $\overline{\mathbf{H}}^{(\mathrm{m})}[n]$ can be written as

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = \overline{\mathbf{Q}}^{(\mathrm{m})}[n]\overline{\boldsymbol{\Sigma}}^{(\mathrm{m})}[n] \left(\overline{\mathbf{F}}^{(\mathrm{m})}[n]\right)^{\mathrm{H}}, \tag{15}$$

where the semi-unitary matrix $\overline{\mathbf{Q}}^{(\mathrm{m})}[n] \in \mathbb{C}^{M_{\mathrm{Rx}}^{(\mathrm{m})} \times \ell_{\max}}$ denotes the matrix of left singular vectors and $\overline{\mathbf{F}}^{(\mathrm{m})}[n] \in \mathbb{C}^{M_{\mathrm{Tx}}^{(\mathrm{m})} \times \ell_{\max}}$ represents the matrix of right singular vectors. The singular value matrix is given by

$$\overline{\boldsymbol{\Sigma}}^{(\mathrm{m})}[n] = \mathrm{diag}\left(\overline{\sigma}_{(1)}^{(\mathrm{m})}[n], \ldots, \overline{\sigma}_{(\ell_{\max})}^{(\mathrm{m})}[n]\right), \tag{16}$$

where the i-th diagonal element $\overline{\sigma}_{(i)}^{(\mathrm{m})}[n]$ of the singular value matrix equals the i-th largest singular value of $\overline{\mathbf{H}}^{(\mathrm{m})}[n]$. Assuming a full-rank channel, the maximum number of streams is $\ell_{\max} = \min\left(M_{\mathrm{Rx}}^{(\mathrm{m})}, M_{\mathrm{Tx}}^{(\mathrm{m})}\right)$. The power loading matrix is given by

$$\overline{\mathbf{P}}^{(\mathrm{m})}[n] = \mathrm{diag}\left(\overline{p}_{(1)}^{(\mathrm{m})}[n], \ldots, \overline{p}_{(\ell_{\max})}^{(\mathrm{m})}[n]\right). \tag{17}$$

To achieve the highest transmission rate, the diagonal elements of $\overline{\mathbf{P}}^{(\mathrm{m})}[n]$ have to be set according to the water-filling power allocation policy. At high SNR, the water-filling power allocation converges to equal power allocation over all streams [22]. Further, the total transmit power constraint

$$\left\|\overline{\mathbf{F}}^{(\mathrm{m})}[n] \left(\overline{\mathbf{P}}^{(\mathrm{m})}[n]\right)^{1/2}\right\|_F^2 = P_{\mathrm{T}}^{(\mathrm{m})} \tag{18}$$

is satisfied by the precoder for all subcarriers.

The symbol vector to be transmitted is written as $\mathbf{x}^{(\mathrm{m})}[n] \in \mathbb{C}^{\ell_{\max}\times 1}$. Prior to transmission over the wireless channel, the symbol vector $\mathbf{x}^{(\mathrm{m})}[n]$ is precoded with a precoding matrix $\overline{\mathbf{F}}^{(\mathrm{m})}[n]$. At the receiver, combining is performed with a combining matrix $\left(\overline{\mathbf{Q}}^{(\mathrm{m})}[n]\right)^{\mathrm{H}}$. With this notation, the input-output relationship for the data transmission phase is then given by

$$\begin{aligned}\mathbf{y}^{(\mathrm{m})}[n] = &\left(\overline{\mathbf{Q}}^{(\mathrm{m})}[n]\right)^{\mathrm{H}} \mathbf{H}^{(\mathrm{m})}[n]\overline{\mathbf{F}}^{(\mathrm{m})}[n]\left(\overline{\mathbf{P}}^{(\mathrm{m})}[n]\right)^{1/2}\mathbf{x}^{(\mathrm{m})}[n] \\ &+ \left(\overline{\mathbf{Q}}^{(\mathrm{m})}[n]\right)^{\mathrm{H}}\mathbf{w}^{(\mathrm{m})}[n],\end{aligned} \tag{19}$$

where the received signal is denoted by $\mathbf{y}^{(\mathrm{m})}[n] \in \mathbb{C}^{\ell_{\max}\times 1}$ and the AWGN added at the mmWave receiver is denoted by $\mathbf{w}^{(\mathrm{m})}[n] \in \mathcal{CN}(0, \sigma^2_{w^{(\mathrm{m})}}\mathbf{I}_{M^{(\mathrm{m})}_{\mathrm{Rx}}})$. Furthermore, since $\overline{\mathbf{Q}}^{(\mathrm{m})}[n]$ is a semi-unitary matrix, $\left(\overline{\mathbf{Q}}^{(\mathrm{m})}[n]\right)^{\mathrm{H}}\mathbf{w}^{(\mathrm{m})}[n]$ follows the same distribution as $\mathbf{w}^{(\mathrm{m})}[n]$.

III. In-Band Channel Estimation Methods

In this section, we revisit the baseline in-band channel estimation methods for the mmWave system. Firstly, we explain the conventional method from Section II-B1, followed by the orthogonal matching pursuit (OMP) algorithm which improves the quality of channel estimation utilizing sparse channel recovery.

A. Conventional Method

This method represents the most straightforward approach for mmWave channel estimation. It relies solely on the in-band mmWave channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$ obtained via LS estimation at pilot positions, followed by linear interpolation, as detailed in Section II-B1. The resulting mmWave channel estimate $\overline{\mathbf{H}}^{(\mathrm{m})}[n]$ for the conventional channel estimation method is given by

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = \widetilde{\mathbf{H}}^{(\mathrm{m})}[n]. \tag{20}$$

Note that this approach does not leverage any additional side information or advanced signal processing techniques.

B. Orthogonal Matching Pursuit (OMP)

To improve the performance of the conventional method, the authors in [52] propose an efficient open-loop channel estimation technique for mmWave MIMO systems based on the OMP algorithm [53]. This method formulates the channel estimation problem as a sparse signal recovery task, leveraging the sparse nature of mmWave channels. In applying the OMP algorithm, a set of quantized angular parameters, referred to as grids, is defined as $\mathcal{G} = \{\vartheta_g, \varphi_g : \vartheta_g, \varphi_g \in [0,\pi],\ g \in \{1,\ldots,G\}\}$. The grids ϑ_g and φ_g are chosen to satisfy $\cos(\vartheta_g) = \cos(\varphi_g) = \frac{2}{G}(g-1) - 1$, where $G \geq \max\{M^{(\mathrm{m})}_{\mathrm{Tx}}, M^{(\mathrm{m})}_{\mathrm{Rx}}\}$. The grid points are non-uniformly distributed in the angular range $[0,\pi]$. By assembling all the array response vectors, the array response matrices defined as

$$\bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Tx}} = \left[\mathbf{a}^{(\mathrm{m})}_{\mathrm{Tx}}(\vartheta_1) \quad \cdots \quad \mathbf{a}^{(\mathrm{m})}_{\mathrm{Tx}}(\vartheta_G)\right] \tag{21}$$

and

$$\bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Rx}} = \left[\mathbf{a}^{(\mathrm{m})}_{\mathrm{Rx}}(\varphi_1) \quad \cdots \quad \mathbf{a}^{(\mathrm{m})}_{\mathrm{Rx}}(\varphi_G)\right] \tag{22}$$

serve as the dictionary in the compressed sensing (CS) formulation. Using these matrices, the channel matrix $\mathbf{H}^{(\mathrm{m})}[n]$ can be expressed as:

$$\mathbf{H}^{(\mathrm{m})}[n] = \bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Rx}}\,\mathbf{H}^{(\mathrm{m})}_a[n]\left(\bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Tx}}\right)^{\mathrm{H}} + \mathbf{E}^{(\mathrm{m})}[n] \tag{23}$$

where $\bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Rx}}\,\mathbf{H}^{(\mathrm{m})}_a[n]\left(\bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Tx}}\right)^{\mathrm{H}}$ represents the quantized channel and $\mathbf{E}^{(\mathrm{m})}[n]$ denotes the quantization error matrix. Since $\mathbf{H}^{(\mathrm{m})}_a[n] \in \mathbb{C}^{G\times G}$ is an L-sparse matrix with L non-zero entries corresponding to the AoDs/AoAs, the channel estimation problem can be formulated as a CS problem:

$$\mathbf{y}^{(\mathrm{m})}[n] = \mathbf{\Psi}^{(\mathrm{m})}\mathbf{h}_a{}^{(\mathrm{m})}[n] + \mathbf{e}^{(\mathrm{m})}[n] + \mathbf{w}^{(\mathrm{m})}[n] \tag{24}$$

where $\mathbf{h}_a{}^{(\mathrm{m})}[n] = \mathrm{vec}\left(\mathbf{H}^{(\mathrm{m})}_a[n]\right)$,

$$\mathbf{\Psi}^{(\mathrm{m})} = \left(\mathbf{U}^{(\mathrm{m})}_{\mathrm{Tx}}\right)^{\mathrm{T}}\left(\bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Tx}}\right)^{*} \otimes \left(\mathbf{U}^{(\mathrm{m})}_{\mathrm{Rx}}\right)^{\mathrm{H}}\bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Rx}} \tag{25}$$

represents the equivalent sensing matrix, and

$$\mathbf{e}^{(\mathrm{m})}[n] = \left(\left(\mathbf{U}^{(\mathrm{m})}_{\mathrm{Tx}}\right)^{\mathrm{T}} \otimes \left(\mathbf{U}^{(\mathrm{m})}_{\mathrm{Rx}}\right)^{\mathrm{H}}\right)\mathrm{vec}\left(\mathbf{E}^{(\mathrm{m})}[n]\right) \tag{26}$$

is the vectorized form of $\mathbf{E}^{(\mathrm{m})}[n]$. In Eq. (25) and Eq. (26), the matrices $\mathbf{U}^{(\mathrm{m})}_{\mathrm{Tx}} \in \mathbb{C}^{M^{(\mathrm{m})}_{\mathrm{Tx}}\times M^{(\mathrm{m})}_{\mathrm{Tx}}}$ and $\mathbf{U}^{(\mathrm{m})}_{\mathrm{Rx}} \in \mathbb{C}^{M^{(\mathrm{m})}_{\mathrm{Rx}}\times M^{(\mathrm{m})}_{\mathrm{Rx}}}$ are DFT matrices normalized by scalars $\sqrt{M^{(\mathrm{m})}_{\mathrm{Tx}}}$ and $\sqrt{M^{(\mathrm{m})}_{\mathrm{Rx}}}$, respectively. The observations $\mathbf{y}^{(\mathrm{m})}[n]$ are obtained by transforming the conventional in-band channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$ into the angular domain, represented as

$$\mathbf{y}^{(\mathrm{m})}[n] = \mathrm{vec}\left(\left(\mathbf{U}^{(\mathrm{m})}_{\mathrm{Rx}}\right)^{\mathrm{H}}\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]\,\mathbf{U}^{(\mathrm{m})}_{\mathrm{Tx}}\right). \tag{27}$$

The optimization problem for CS-based channel estimation can then be formulated as:

$$\begin{aligned}\overline{\mathbf{h}}^{(\mathrm{m})}_a[n] = \arg\min_{\mathbf{h}_a} &\left\|\mathbf{y}^{(\mathrm{m})}[n] - \mathbf{\Psi}^{(\mathrm{m})}\mathbf{h}^{(\mathrm{m})}_a[n]\right\|_2 \\ \text{subject to} \quad &\left\|\mathbf{h}^{(\mathrm{m})}_a[n]\right\|_0 = L\end{aligned} \tag{28}$$

where $\overline{\mathbf{h}}^{(\mathrm{m})}_a[n]$ denotes the CS-based estimate of $\mathbf{h}^{(\mathrm{m})}_a[n]$. The obtained vectorized estimate is then reshaped back to matrix form, yielding $\overline{\mathbf{H}}^{(\mathrm{m})}_a = \mathrm{vec}^{-1}\left(\overline{\mathbf{h}}_a\right)$, and the desired quantized channel estimate $\overline{\mathbf{H}}^{(\mathrm{m})}$ is given by

$$\overline{\mathbf{H}}^{(\mathrm{m})} = \bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Rx}}\overline{\mathbf{H}}^{(\mathrm{m})}_a\left(\bar{\mathbf{A}}^{(\mathrm{m})}_{\mathrm{Tx}}\right)^{\mathrm{H}}. \tag{29}$$

Further details on the procedure and the OMP algorithm can be found in [52].

IV. Joint AoD and AoA Estimation

Our training phase (see Section II-B1) relies on the AoD and AoA between the transmitter and receiver, parameters that are often unknown in wireless communications. To address this, we employ the Bartlett beamformer in this section for joint estimation of AoD and AoA for the LOS component in the sub-6 GHz band. To simplify the notation, we treat the 2D case

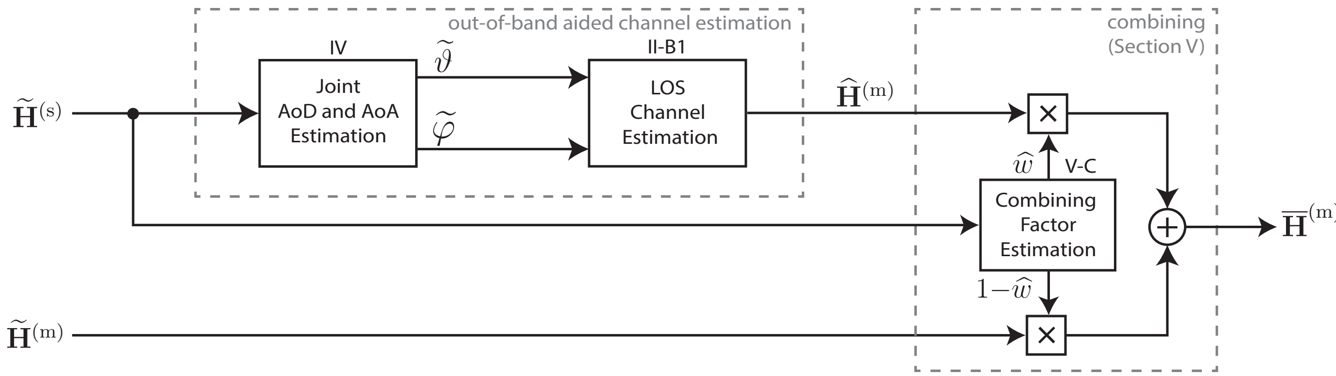

Fig. 4. We apply the proposed out-of-band aided channel estimation approach to obtain the mmWave LOS channel estimate. We obtained the improved mmWave channel estimate $\overline{\mathbf{H}}^{(\mathrm{m})}$ by combining the out-of-band aided LOS estimate $\widehat{\mathbf{H}}^{(\mathrm{m})}$ and the in-band mmWave estimate $\widetilde{\mathbf{H}}^{(\mathrm{m})}$.

here. The approach can be easily extended to 3D by adding elevation estimation into the algorithm.

Directly estimating AoD and AoA in the mmWave band using only the channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$ is challenging. This challenge occurs due to the lack of beamforming at the transmitter and significant propagation losses inherent to mmWave systems. During channel estimation, the transmitter can only transmit isotropically, resulting in a low pre-beamforming SNR. Consequently, this low pre-beamforming SNR leads to a highly imprecise mmWave channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$. In contrast, due to smaller propagation losses and smaller bandwidth, a sub-6 GHz system has an $\alpha\beta$ times higher SNR (see Eq. (7)) compared to the corresponding mmWave system (see Section II-A). This higher SNR allows for a more accurate channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{s})}[n]$ and consequently more accurate estimation of AoD and AoA.

Therefore, we jointly estimate the AoD and AoA using the 2D-Bartlett beamformer [45] within the sub-6 GHz system. To enable joint AoD and AoA estimation, we convert the sub-6 GHz channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{s})}[n]$ into its vectorized form:

$$\widetilde{\mathbf{h}}^{(\mathrm{s})}[n] = \mathrm{vec}\left(\widetilde{\mathbf{H}}^{(\mathrm{s})}[n]\right) \in \mathbb{C}^{M_{\mathrm{Tx}}^{(\mathrm{s})} M_{\mathrm{Rx}}^{(\mathrm{s})} \times 1} \tag{30}$$

Since our goal is to estimate the AoA and AoD for the LOS channel, we focus solely on the zero-delay channel tap and suppress all other taps, as they do not contain LOS information. To achieve this, we first obtain the delay-domain vectorized channel estimate $\widetilde{\mathbf{h}}^{(\mathrm{s})}[\tau]$ by applying the IDFT to $\widetilde{\mathbf{h}}^{(\mathrm{s})}[n]$. We then filter this estimate to isolate the LOS component:

$$\widetilde{\mathbf{h}}_{\mathrm{LOS}}^{(\mathrm{s})}[\tau] = \widetilde{\mathbf{h}}^{(\mathrm{s})}[\tau]\delta[\tau] \tag{31}$$

Note that this LOS isolation may be imperfect and could include some residual unfiltered channel components. Subsequently, we convert the LOS channel estimate back to the frequency domain by applying the DFT to $\widetilde{\mathbf{h}}_{\mathrm{LOS}}^{(\mathrm{s})}[\tau]$. Prior to the angular spectrum calculation, we estimate the covariance matrix as

$$\widetilde{\mathbf{C}}^{(\mathrm{s})} = \widetilde{\mathbf{h}}_{\mathrm{LOS}}^{(\mathrm{s})}[n]\left(\widetilde{\mathbf{h}}_{\mathrm{LOS}}^{(\mathrm{s})}[n]\right)^{\mathrm{H}} \tag{32}$$

using a single subcarrier $n = 1$ due to the frequency-flat nature of $\widetilde{\mathbf{h}}_{\mathrm{LOS}}^{(\mathrm{s})}[n]$. The angular power spectrum of the 2D-Bartlett beamformer is obtained by computing

$$P_{\vartheta,\varphi}^{(\mathrm{s})}(\vartheta_{\mathrm{tr}}, \varphi_{\mathrm{tr}}) = \frac{\left(\mathbf{a}^{(\mathrm{s})}(\vartheta_{\mathrm{tr}}, \varphi_{\mathrm{tr}})\right)^{\mathrm{H}} \widetilde{\mathbf{C}}^{(\mathrm{s})}\, \mathbf{a}^{(\mathrm{s})}(\vartheta_{\mathrm{tr}}, \varphi_{\mathrm{tr}})}{\left(\mathbf{a}^{(\mathrm{s})}(\vartheta_{\mathrm{tr}}, \varphi_{\mathrm{tr}})\right)^{\mathrm{H}} \mathbf{a}^{(\mathrm{s})}(\vartheta_{\mathrm{tr}}, \varphi_{\mathrm{tr}})}, \tag{33}$$

where

$$\mathbf{a}^{(\mathrm{s})}(\vartheta_{\mathrm{tr}}, \varphi_{\mathrm{tr}}) = \left(\mathbf{a}_{\mathrm{Tx}}^{(\mathrm{s})}(\vartheta_{\mathrm{tr}})\right)^{*} \otimes \mathbf{a}_{\mathrm{Rx}}^{(\mathrm{s})}(\varphi_{\mathrm{tr}}) \tag{34}$$

is the joint steering vector for the trial angles ϑ_{tr} and φ_{tr}. The trial angles ϑ_{tr} and φ_{tr} can each take $L_{\vartheta_{\mathrm{tr}}}$ and $L_{\varphi_{\mathrm{tr}}}$ potential angles within the given range. Finally, the estimated AoD $\widetilde{\vartheta}$ and AoA $\widetilde{\varphi}$ are found through the maxima of the angular power spectrum

$$\widetilde{\vartheta}, \widetilde{\varphi} = \underset{\vartheta_{\mathrm{tr}}, \varphi_{\mathrm{tr}}}{\mathrm{argmax}} \left\{ P_{\vartheta,\varphi}^{(\mathrm{s})}(\vartheta_{\mathrm{tr}}, \varphi_{\mathrm{tr}}) \right\}. \tag{35}$$

V. Out-of-Band Aided Channel Estimation Methods

In this section, we propose three channel estimation methods for the mmWave system. Having both, in-band $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$ and out-of-band aided $\widehat{\mathbf{H}}^{(\mathrm{m})}[n]$ channel estimates available, we combine them to improve the quality of channel estimation. The approach is detailed in the block diagram shown in Fig. 4.

A. Translating

A most straightforward approach to utilize out-of-band information for establishing the mmWave link is to only use the LOS channel estimate $\widehat{\mathbf{H}}^{(\mathrm{m})}[n]$ from Eq. (14) as follows

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = \widehat{\mathbf{H}}^{(\mathrm{m})}[n]. \tag{36}$$

As the K-factor increases, the amplitude of the stochastic Rayleigh component diminishes and its impact on the performance becomes negligible. This enhances the prominence of the deterministic LOS component, allowing for a more accurate estimation of the mmWave channel using solely the estimated mmWave LOS channel $\widehat{\mathbf{H}}^{(\mathrm{m})}[n]$. For low K-factors, where the stochastic component dominates the deterministic one, there is no benefit to be gained from using this method (actually, this method performs worse than the conventional one, as demonstrated in Section VI).

B. Averaging

To enhance performance for lower K-factors, we propose a method that incorporates the out-of-band aided channel estimate $\widehat{\mathbf{H}}^{(\mathrm{m})}[n]$ along with the in-band mmWave channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$. The resulting channel estimate is then given by the average

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = \frac{\widehat{\mathbf{H}}^{(\mathrm{m})}[n] + \widetilde{\mathbf{H}}^{(\mathrm{m})}[n]}{2}. \tag{37}$$

At moderate K-factors, this method yields better results on average than the translating method. However, this method involves one more implementation step compared to the translating method. In cases of low and high K-factors, there is still room for improvement by assigning weights other than 0.5 to $\widehat{\mathbf{H}}^{(\mathrm{m})}[n]$ and $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$ in Eq. (37).

C. Maximal Ratio Combining (MRC)

Further performance improvement is achieved through optimal combining of the out-of-band aided $\widehat{\mathbf{H}}^{(\mathrm{m})}[n]$ and in-band $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$ channel estimates. Since the wireless channel is changing, it is important to select an optimum weighting factors for each specific combination of the mmWave K-factor $K^{(\mathrm{m})}$ and the mmWave SNR $\gamma^{(\mathrm{m})}$. This optimal combination strategy is achieved through MRC of the out-of-band and in-band channel estimates, expressed as:

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = \widehat{w}\,\widehat{\mathbf{H}}^{(\mathrm{m})}[n] + (1-\widehat{w})\widetilde{\mathbf{H}}^{(\mathrm{m})}[n] \tag{38}$$

The approximation of the optimal weight $\widehat{w}$ is derived as follows. Since $\widehat{w}$ is independent of any specific subcarrier, we omit the subcarrier index n in the subsequent derivation. Based on Eq. (12) and Eq. (14), the out-of-band aided channel estimate can be expressed as

$$\begin{aligned}\widehat{\mathbf{H}}^{(\mathrm{m})} = &\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta)\\ &+ \underbrace{\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{W}^{(\mathrm{m})}\left(\mathbf{a}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta)\right)^{\mathrm{H}}}_{c_{\mathrm{W}}^{(\mathrm{m})}\mathbf{W}^{(\mathrm{m})}},\end{aligned} \tag{39}$$

where $\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi) = \mathbf{a}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\left(\mathbf{a}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\right)^{\mathrm{H}}$ and $\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta) = \mathbf{a}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta)\left(\mathbf{a}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta)\right)^{\mathrm{H}}$ project the channel and the AWGN onto the space spanned by $\mathbf{a}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)$ and $\mathbf{a}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta)$. The AWGN is denoted by $\mathbf{W}^{(\mathrm{m})} \in \mathcal{CN}\left(\mathbf{0}, M_{\mathrm{Rx}}^{(\mathrm{m})}\sigma_{w^{(\mathrm{m})}}^2\mathbf{I}_{M_{\mathrm{Tx}}^{(\mathrm{m})}}\right)$. The noise scaling factor is given by $c_{\mathrm{W}}^{(\mathrm{m})} = 1/\sqrt{M_{\mathrm{Tx}}^{(\mathrm{m})}M_{\mathrm{Rx}}^{(\mathrm{m})}}$. Next, the in-band channel estimate is represented by

$$\widetilde{\mathbf{H}}^{(\mathrm{m})} = \mathbf{H}^{(\mathrm{m})} + \mathbf{W}^{(\mathrm{m})}. \tag{40}$$

We assume that $\mathbf{W}^{(\mathrm{m})}$ and $\mathbf{H}^{(\mathrm{m})}$ are uncorrelated. We seek the real-valued scalar $\widehat{w} > 0$ that minimizes the MSE, i.e.,

$$\widehat{w} = \arg\min_{w} \mathbb{E}\left\{\left\|w\widehat{\mathbf{H}}^{(\mathrm{m})} + (1-w)\widetilde{\mathbf{H}}^{(\mathrm{m})} - \mathbf{H}^{(\mathrm{m})}\right\|_F^2\right\}. \tag{41}$$

The expectation $\mathbb{E}\{\cdot\}$ from Eq. (41) with respect to the channel $\mathbf{H}^{(\mathrm{m})}$ is given by

$$\begin{aligned}&\mathbb{E}\left\{\left\|w\widehat{\mathbf{H}}^{(\mathrm{m})} + (1-w)\widetilde{\mathbf{H}}^{(\mathrm{m})} - \mathbf{H}^{(\mathrm{m})}\right\|_F^2\right\}\\ &= w^2\mathbb{E}\left\{\left\|\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta) - \mathbf{H}^{(\mathrm{m})}\right\|_F^2\right\}\\ &\quad + \left[\left(wc_{\mathrm{W}}^{(\mathrm{m})}\right)^2 + (1-w)^2\right]\underbrace{\mathbb{E}\left\{\left\|\mathbf{W}^{(\mathrm{m})}\right\|_F^2\right\}}_{M_{\mathrm{Tx}}^{(\mathrm{m})}M_{\mathrm{Rx}}^{(\mathrm{m})}\sigma_{w^{(\mathrm{m})}}^2}.\end{aligned} \tag{42}$$

Next, we set the derivative equal to zero, resulting in

$$\begin{aligned}&\frac{\partial}{\partial w}\left(w^2\mathbb{E}\left\{\left\|\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta) - \mathbf{H}^{(\mathrm{m})}\right\|_F^2\right\}\right.\\ &\left.+ w^2\sigma_{w^{(\mathrm{m})}}^2 + (1-w)^2 M_{\mathrm{Tx}}^{(\mathrm{m})}M_{\mathrm{Rx}}^{(\mathrm{m})}\sigma_{w^{(\mathrm{m})}}^2\right) = 0,\end{aligned} \tag{43}$$

which gives

$$\begin{aligned}&2w\,\underbrace{\mathbb{E}\left\{\left\|\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta) - \mathbf{H}^{(\mathrm{m})}\right\|_F^2\right\}}_{\Gamma(\vartheta,\varphi)}\\ &+ 2\left(w + wM_{\mathrm{Tx}}^{(\mathrm{m})}M_{\mathrm{Rx}}^{(\mathrm{m})} - M_{\mathrm{Tx}}^{(\mathrm{m})}M_{\mathrm{Rx}}^{(\mathrm{m})}\right)\sigma_{w^{(\mathrm{m})}}^2 = 0.\end{aligned} \tag{44}$$

Hence, the optimal $\widehat{w}$ is given by

$$\widehat{w} = \frac{M_{\mathrm{Tx}}^{(\mathrm{m})}M_{\mathrm{Rx}}^{(\mathrm{m})}\sigma_{w^{(\mathrm{m})}}^2}{\Gamma(\vartheta,\varphi) + \left(1 + M_{\mathrm{Tx}}^{(\mathrm{m})}M_{\mathrm{Rx}}^{(\mathrm{m})}\right)\sigma_{w^{(\mathrm{m})}}^2}. \tag{45}$$

The matrices $\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)$ and $\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta)$ project the channel $\mathbf{H}^{(\mathrm{m})}$ such that the Rayleigh channel component is filtered out, i.e., $\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}_{\mathrm{fs}}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta) \approx \mathbf{H}_{\mathrm{fs}}^{(\mathrm{m})}$ and $\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}_{\mathrm{rp}}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta) \approx \mathbf{0}$. Hence, the expectation $\Gamma(\vartheta,\varphi)$ from Eq. (44) is approximated by

$$\begin{aligned}\Gamma(\vartheta,\varphi) &= \mathbb{E}\left\{\left\|\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta) - \mathbf{H}^{(\mathrm{m})}\right\|_F^2\right\}\\ &= \mathbb{E}\left\{\left\|\sqrt{\frac{K^{(\mathrm{m})}}{1+K^{(\mathrm{m})}}}\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}_{\mathrm{fs}}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta)\right.\right.\\ &\quad + \sqrt{\frac{1}{1+K^{(\mathrm{m})}}}\mathbf{A}_{\mathrm{Rx}}^{(\mathrm{m})}(\varphi)\,\mathbf{H}_{\mathrm{rp}}^{(\mathrm{m})}\mathbf{A}_{\mathrm{Tx}}^{(\mathrm{m})}(\vartheta)\\ &\quad \left.\left. - \left(\sqrt{\frac{K^{(\mathrm{m})}}{1+K^{(\mathrm{m})}}}\mathbf{H}_{\mathrm{fs}}^{(\mathrm{m})} + \sqrt{\frac{1}{1+K^{(\mathrm{m})}}}\mathbf{H}_{\mathrm{rp}}^{(\mathrm{m})}\right)\right\|_F^2\right\}\\ &\approx \frac{1}{1+K^{(\mathrm{m})}}\mathbb{E}\left\{\left\|\mathbf{H}_{\mathrm{rp}}^{(\mathrm{m})}\right\|_F^2\right\} \approx \frac{M_{\mathrm{Tx}}^{(\mathrm{m})}M_{\mathrm{Rx}}^{(\mathrm{m})}}{1+K^{(\mathrm{m})}}.\end{aligned} \tag{46}$$

As the K-factor is unknown at the receiver, it needs to be estimated. Directly estimating the K-factor in the mmWave band using only the channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{m})}[n]$ is challenging due to the absence of beamforming at the transmitter and significant propagation losses inherent to mmWave systems. Thus, we estimate the K-factor in the sub-6 GHz band using the channel estimate $\widetilde{\mathbf{H}}^{(\mathrm{s})}[n]$ and the K-factor estimation method based on the method of moments [54], [55]. Finally, with the estimated K-factor $\widetilde{K}^{(\mathrm{s})}$ available, the optimal $\widehat{w}$ can be approximated by

$$\widehat{w} \approx \frac{M_{\mathrm{Tx}}^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})} \sigma_{w^{(\mathrm{m})}}^{2}}{\frac{M_{\mathrm{Tx}}^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})}}{1+c_K \widetilde{K}^{(\mathrm{s})}} + \left(1 + M_{\mathrm{Tx}}^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})}\right) \sigma_{w^{(\mathrm{m})}}^{2}}. \tag{47}$$

VI. Simulation-based Comparison

In this section, we investigate the performance of the proposed channel estimation methods by means of Monte-Carlo simulations. Firstly, we introduce the performance metrics in Section VI-A. Following that, we validate the performance of the proposed methods as a function of the K-factor in Section VI-B and the SNR in Section VI-C. Moreover, we explore the impact of varying the numbers of transmit and receive antennas in Section VI-D, while Section VI-E examines the impact of the scaling factor c_K. Finally, we evaluate the computational complexity of the proposed methods in Section VI-F and analyze the overhead in Section VI-G.

A. Performance Metrics

To evaluate the proposed channel estimation methods, we simulate the normalized channel estimation and interpolation MSE and the achievable SE in a frequency-selective channel. The simulation parameters are summarized in Table I. Unless specified otherwise, we assume equal numbers of antennas for both sub-6 GHz and mmWave systems $\left(M_{\mathrm{Tx}}^{(\mathrm{s})} = M_{\mathrm{Tx}}^{(\mathrm{m})} \text{ and } M_{\mathrm{Rx}}^{(\mathrm{s})} = M_{\mathrm{Rx}}^{(\mathrm{m})}\right)$, though different antenna configurations are also possible.

For the frequency-selective channel model, we utilize the 3GPP model for the urban macro LOS scenario with an adjustable K-factor, following the procedure outlined in [49]. The multi-band MIMO measurements presented in Section VII provide an estimated sub-6 GHz K-factor of 2.91 dB and an estimated mmWave K-factor of 2.35 dB, calculated using the K-factor estimation method outlined in [54]. Given the close similarity between K-factors [56], we set the scaling factor $c_K = 1$ ($c_K = 0$ dB), thereby defining the mmWave K-factor as $K^{(\mathrm{m})} = K^{(\mathrm{s})}$. Additionally, we assume that the AoD and AoA are mutually independent and uniformly distributed within the range $[-90°, 90°]$.

As a baseline, we first evaluate the performance of conventional channel estimation using only the mmWave channel estimate (see Section III-A). Additionally, we present results for the sparse channel estimation method based on the OMP algorithm (see Section III-B). Furthermore, we show the performance under perfect CSI conditions

$$\overline{\mathbf{H}}^{(\mathrm{m})}[n] = \mathbf{H}^{(\mathrm{m})}[n]. \tag{48}$$

1) Normalized Channel Mean Squared Error: The normalized channel estimation and interpolation MSE is given by

$$\mathrm{NMSE} = \frac{1}{L_r N^{(\mathrm{m})}} \sum_{l=1}^{L_r} \sum_{n=1}^{N^{(\mathrm{m})}} \frac{\left\| \mathbf{H}_l^{(\mathrm{m})}[n] - \overline{\mathbf{H}}_l^{(\mathrm{m})}[n] \right\|_F^2}{\left\| \mathbf{H}_l^{(\mathrm{m})}[n] \right\|_F^2}, \tag{49}$$

where l denotes a specific channel realization.

TABLE I
Simulation Parameters

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing $\triangle f$	60 kHz	120 kHz
Cyclic Prefix t_{CP}	1.19 μs	0.59 μs
Transmit Power P_T	30 dBm	30 dBm
Noise Figure F	3 dB	3 dB
Number of Realizations L_r	1000	1000

2) Root Mean Squared Error of Angles: We evaluate the performance of the utilized joint AoD/AoA estimation for the LOS component using the root mean squared error (RMSE) for both the AoD and AoA. The AoD RMSE is given by

$$\mathrm{RMSE}_{\vartheta} = \sqrt{\frac{1}{L_r} \sum_{l=1}^{L_r} \left| \vartheta_l - \widetilde{\vartheta}_l \right|^2}, \tag{50}$$

while the AoA RMSE, denoted as RMSE_{φ}, is defined analogously by replacing ϑ_l and $\widetilde{\vartheta}_l$ with φ_l and $\widetilde{\varphi}_l$, respectively. In Eq. (50), ϑ_l represents the true AoD for the l-th channel realization, and $\widetilde{\vartheta}_l$ denotes its corresponding estimate.

3) Achievable Spectral Efficiency: We consider the achievable SE (achievable rate per bandwidth) as the main performance metric for MIMO systems. Since the SE is rate per bandwidth, this metric does not depend on the employed transmission bandwidth and therefore facilitates direct comparison of results. The achievable SE in bits/s/Hz averaged over $N^{(\mathrm{m})}$ subcarriers is given by

$$\mathrm{SE} = \frac{1}{N^{(\mathrm{m})}} \sum_{n=1}^{N^{(\mathrm{m})}} \sum_{\mu=1}^{\ell_{\max}} \log_2 \left(1 + \mathrm{SINR}_{\mu}[n]\right) \tag{51}$$

with the effective signal-to-interference-and-noise ratio (SINR) for the stream μ denoted by [57]

$$\mathrm{SINR}_{\mu}[n] = \frac{\left| \overline{\mathrm{G}}_{\mu,\mu}^{(\mathrm{m})}[n] \right|^2}{\sum\limits_{\substack{\nu=1 \\ \nu \neq \mu}}^{\ell_{\max}} \left| \overline{\mathrm{G}}_{\mu,\nu}^{(\mathrm{m})}[n] \right|^2 + \left(\sigma_{\mu}^2[n] + \sigma_{w^{(\mathrm{m})}}^2\right) \left\| \overline{\mathrm{Q}}_{:,\mu}^{(\mathrm{m})}[n] \right\|^2}. \tag{52}$$

In Eq. (52), $\overline{\mathrm{G}}_{\mu,\nu}^{(\mathrm{m})}[n]$ with $\mu, \nu \in \{1, \ldots, \ell_{\max}\}$ represent entries of the channel gain matrix $\overline{\mathbf{G}}^{(\mathrm{m})}[n] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$ for the n-th subcarrier, which is given by

$$\overline{\mathbf{G}}^{(\mathrm{m})}[n] = \left(\overline{\mathbf{Q}}^{(\mathrm{m})}[n]\right)^{\mathrm{H}} \mathbf{H}^{(\mathrm{m})}[n] \overline{\mathbf{F}}^{(\mathrm{m})}[n] \left(\overline{\mathbf{P}}^{(\mathrm{m})}[n]\right)^{1/2}. \tag{53}$$

Further, $\sigma_{\mu}^2[n]$ represents the diagonal entries (variance) of the estimation error covariance matrix $\overline{\mathbf{C}}_{\varepsilon}^{(\mathrm{m})}[n] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$ for the n-th subcarrier, given by

$$\overline{\mathbf{C}}_{\varepsilon}^{(\mathrm{m})}[n] = \frac{1}{\ell_{\max}} \overline{\boldsymbol{\varepsilon}}^{(\mathrm{m})}[n] \left(\overline{\boldsymbol{\varepsilon}}^{(\mathrm{m})}[n]\right)^{\mathrm{H}}, \tag{54}$$

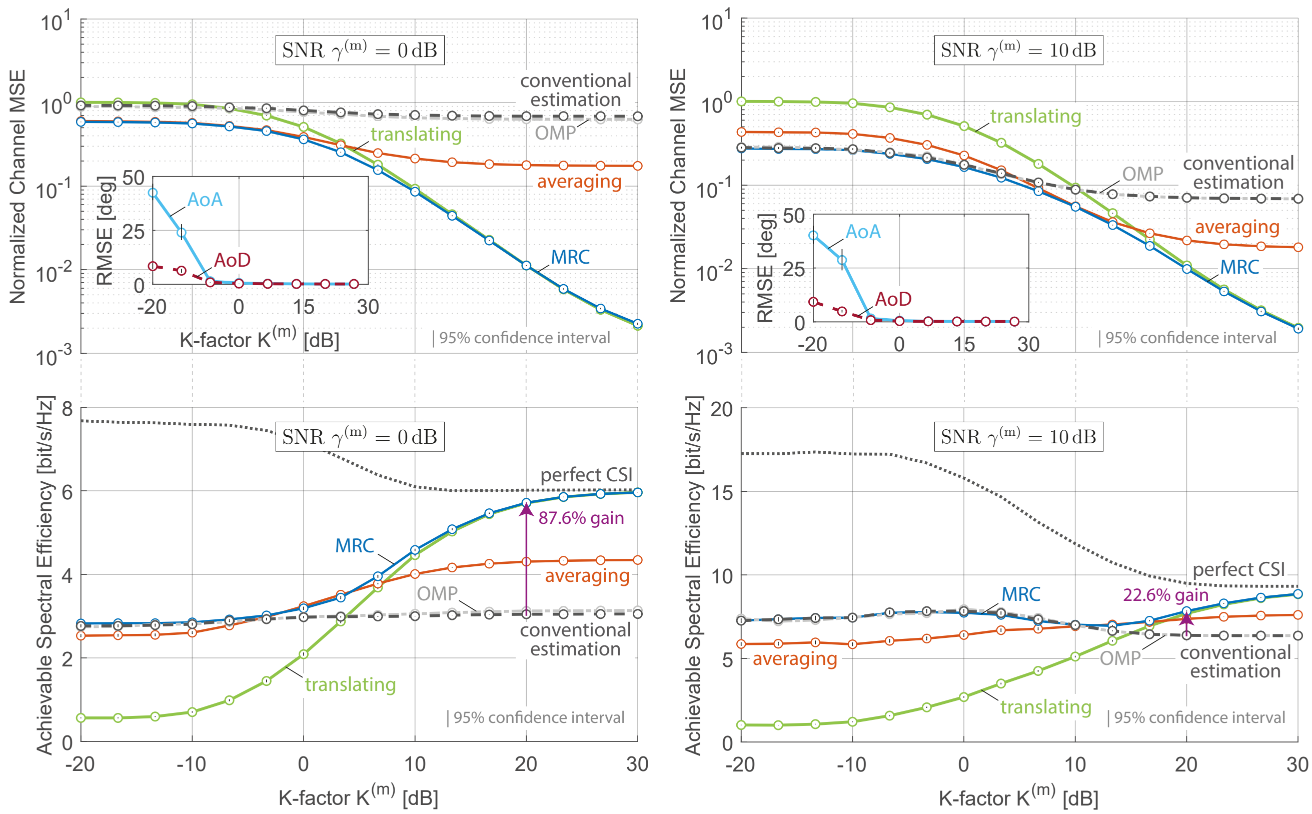

Fig. 5. At lower SNR of 0 dB, the proposed MRC method yields 87.6% gain in achievable SE (left), and at higher SNR of 10 dB the gain is 22.6% (right).

where the estimation error matrix is denoted by

$$\overline{\varepsilon}^{(\mathrm{m})}[n] = \mathbf{G}^{(\mathrm{m})}[n] - \overline{\mathbf{G}}^{(\mathrm{m})}[n]. \quad (55)$$

In Eq. (55), $\overline{\varepsilon}^{(\mathrm{m})}[n]$ represents the error between the channel gain matrix with the estimated precoder and combiner, $\overline{\mathbf{G}}^{(\mathrm{m})}[n]$, and the channel gain matrix with the perfectly designed precoder and combiner $\mathbf{G}^{(\mathrm{m})}[n]$, given by

$$\mathbf{G}^{(\mathrm{m})}[n] = \left(\mathbf{Q}^{(\mathrm{m})}[n]\right)^{\mathrm{H}} \mathbf{H}^{(\mathrm{m})}[n]\mathbf{F}^{(\mathrm{m})}[n] \left(\mathbf{P}^{(\mathrm{m})}[n]\right)^{1/2}. \quad (56)$$

B. Performance as a Function of K-factor

In the first simulation, we investigate the performance of the three proposed channel estimation methods as a function of the K-factor. The simulation results for the 8×8 MIMO configuration are shown in terms of normalized channel MSE and achievable SE in Fig. 5. We further analyze the K-factor values estimated from multi-band (sub-6 GHz and mmWave) measurement campaigns. In the vehicle-to-infrastructure (V2I) scenario reported in [56], the estimated K-factor ranges from -5 to 16 dB. In an urban environment, [19] reports K-factor values mainly between -12 and 30 dB. Additionally, measurements conducted in an outdoor campus environment [58] show K-factor values ranging from -15 to 10 dB. To capture the variability observed across these scenarios, we adopt an extended K-factor range of $[-20, 30]$ dB in our simulations.

We observe that both the channel MSE and the RMSE of the LOS AoA and AoD decrease as the K-factor increases, regardless of the SNR level. For low K-factors, corresponding to NLOS conditions, the AoA RMSE is higher than the AoD RMSE. This occurs due to the presence of numerous scatterers located around the receiver in the urban scenario, which makes accurate AoA estimation more challenging.

Results at SNR of 0 dB: When utilizing the translating method at a high K-factor of 20 dB, it leads to an 87.6% gain in achievable SE compared to relying solely on conventional channel estimation in the mmWave band. However, the performance of the translating method is poor at a low K-factor. The averaging method yields better results at low K-factors, whereas it performs worse than the translating method at high K-factors. Ultimately, the MRC method is for all K-factors at least as good or better than the conventional method. Notably, the improvement in achievable SE becomes apparent from around a K-factor of 0 dB. By calculating the appropriate combining factor $\widehat{w}$, the out-of-band aided and in-band channel estimates are harmoniously combined, ensuring that the resulting achievable SE surpasses or equals the conventional value for each SNR and K-factor. Specifically, for low K-factors, a low combining factor $\widehat{w}$ is chosen to emphasize the in-band mmWave portion dominance. Conversely, for high K-factors, the reverse is true. The OMP-based method provides only a marginal improvement over the conventional method.

Results at SNR of 10 dB: Again, the MRC method consistently outperforms the translating method and the averaging method. However, the gain compared to conventional channel estimation at a high K-factor of 20 dB is now only 22.6%. Furthermore, the improvement becomes discernible primarily

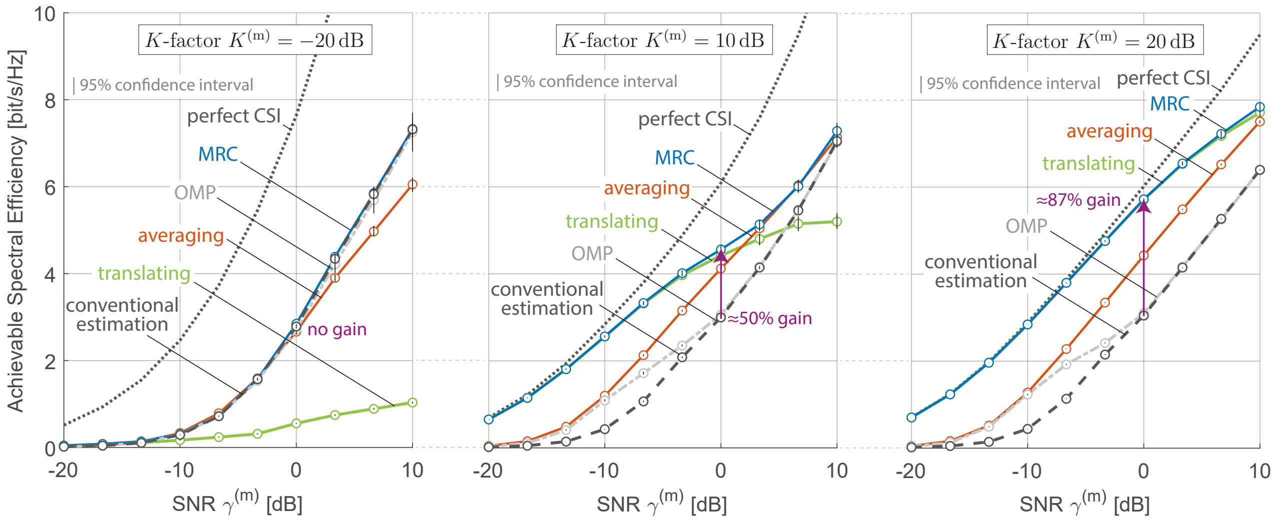

Fig. 6. As K-factor increases, all three proposed channel estimation methods exhibit improvement compared to conventional estimation, gradually approaching the performance achieved with perfect CSI. At a low K-factor of -20 dB, there is no gain in achievable SE (left), at a moderate K-factor of 10 dB, the proposed MRC method yields approximately 50% gain (middle), whereas at a higher K-factor of 20 dB, the gain is approximately 87% (right).

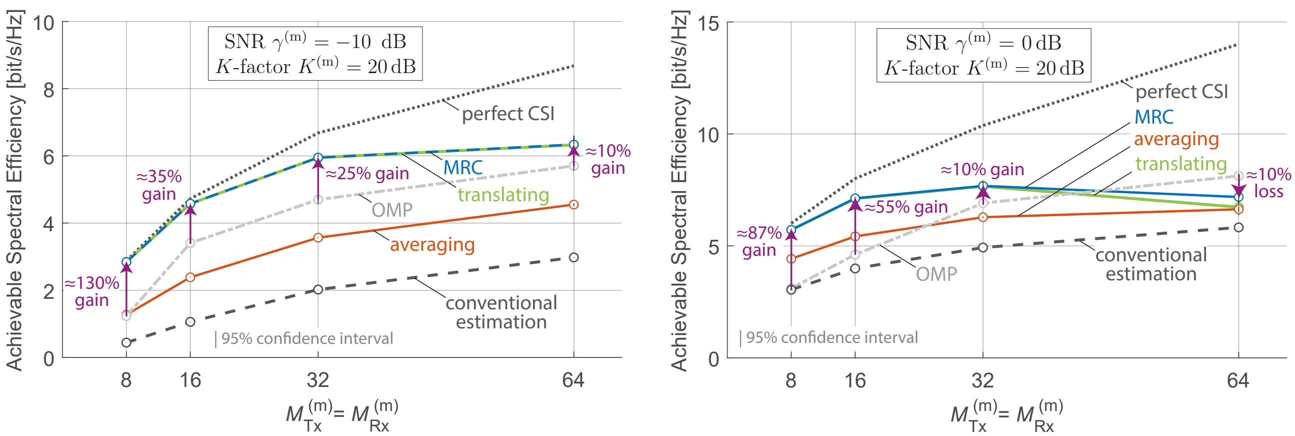

Fig. 7. At lower SNR of -10 dB, improvement is evident irrespective of increased number of mmWave antennas (left), while at higher SNR of 0 dB, notable improvement is obtained with small difference between sub-6 GHz and mmWave antennas (right).

from a K-factor of approximately 10 dB onwards. Here, the performance of the OMP-based method is nearly identical to that of the conventional method.

C. Performance as a Function of SNR

In the next simulation, we investigate the performance as a function of SNR. The simulation results for the 8×8 MIMO configuration in terms of achievable SE are shown in Fig. 6. As anticipated, the performance in terms of achievable SE improves as the SNR increases, regardless of the method used and the K-factor involved.

Results at K-factor of -20 dB: The translating method notably underperforms compared to the conventional method and the OMP-based method across the entire SNR range. Significant improvement is observed with the averaging method, and further improvements are achieved by using the MRC method, matching the performance of the conventional and OMP-based methods.

Results at K-factor of 10 dB: Here, the translating method outperforms both the conventional and OMP-based methods within the SNR range from -20 to 5 dB and outperforms the averaging method within the SNR range from -20 to 0 dB. Furthermore, the MRC method outperforms the conventional and OMP-based methods in terms of achievable SE over the entire SNR range, with a gain of approximately 50% observed at SNR of 0 dB. Within the SNR range of -20 to 0 dB, the OMP-based method surpasses the conventional method, after which their performances converge.

Results at K-factor of 20 dB: In this case, all three proposed channel estimation methods outperform the conventional and OMP-based methods. Of particular note is the close alignment in achievable SE between the translating method and the MRC method, with both methods closely approaching the performance achieved with perfect CSI. Here, we observe a notable gain of approximately 87% at SNR of 0 dB.

D. Performance as a Function of Number of Antennas

In this subsection, we investigate the influence of varying antenna configurations on the performance of the proposed

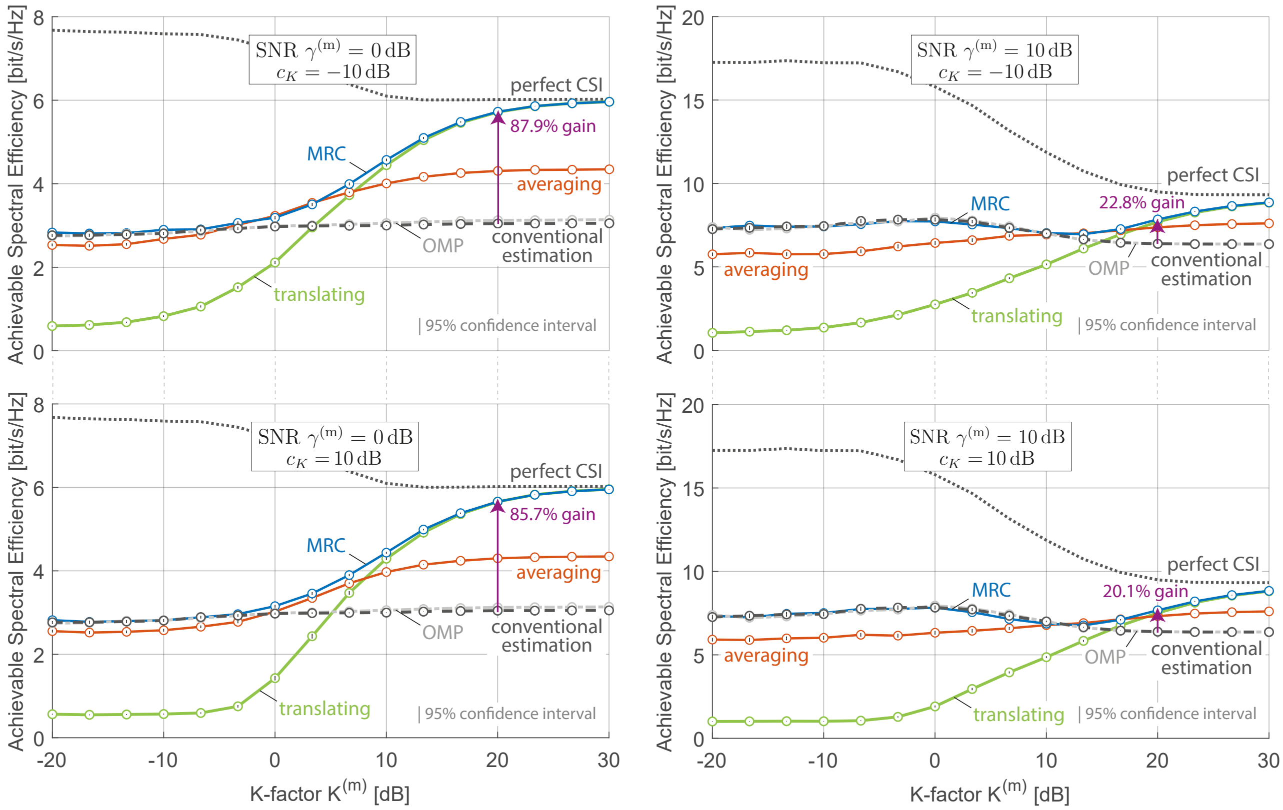

Fig. 8. At a scaling factor c_K of -10 dB (top), the SE gain is 87.9% at 0 dB SNR and 22.8% at 10 dB SNR. When c_K increases to 10 dB (bottom), the SE gain decreases to 85.7% at 0 dB SNR and 20.1% at 10 dB SNR.

methods. The simulation results, expressed in terms of achievable SE, for configurations where $M_{\text{Tx}}^{(\text{s})} = M_{\text{Rx}}^{(\text{s})} = 8$ and $M_{\text{Tx}}^{(\text{m})} = M_{\text{Rx}}^{(\text{m})} \in \{8, 16, 32, 64\}$ with a K-factor of 20 dB, are shown in Fig. 7.

Results at SNR of -10 dB: The translating and MRC methods show comparable performance, both outperforming the averaging, OMP-based and conventional methods. Notably, the OMP-based method performs significantly better than the conventional one in this case. In particular, the achievable SE gain over the OMP-based method decreases as the number of mmWave antennas increases. In the 8×8 MIMO configuration, a substantial SE gain of approximately 130% is observed, while this gain drops to around 10% in the 64×64 configuration. This decrease can be attributed to the increased number of mmWave antennas, which requires highly accurate AoD/AoA estimations. However, such precision cannot be achieved with the smaller sub-6 GHz antenna arrays.

Results at SNR of 0 dB: The performance of the translating method deteriorates due to the greater impact of the resolution mismatch between sub-6 GHz and mmWave systems. The averaging method exhibits moderate performance, whereas the MRC method either matches or outperforms the conventional method across all mmWave antenna configurations. Similar to the results at SNR of -10 dB, the gain in achievable SE decreases as the number of mmWave antennas increases. Additionally, MRC method outperforms the OMP-based method for MIMO configurations of 8×8, 16×16, and 32×32, whereas the OMP-based method surpasses the MRC for the 64×64 configuration. For the 8×8 MIMO configuration, the SE gain over the OMP is approximately 87%, while in the 64×64 configuration, the OMP-based method achieves around 10% better performance than the MRC method.

E. Impact of Scaling Factor c_K

We further evaluate the impact of the scaling factor c_K between the mmWave K-factor $K^{(\text{m})}$ and sub-6 GHz K-factor $K^{(\text{s})}$ (see Section II-A). The simulation results in terms of achievable SE as a function of the mmWave K-factor $K^{(\text{m})}$ are shown in Fig. 8. Overall, the scaling factor c_K has no significant impact on performance. However, minor effects are observed under specific conditions.

Results at scaling factor of -10 dB: Here, the sub-6 GHz K-factor is 10 dB higher than the mmWave K-factor. This enhances the estimation accuracy of the LOS component in the sub-6 GHz band, leading to minor performance improvements in the proposed method. The SE gain over conventional channel estimation at a K-factor of 20 dB increases to 87.9% at 0 dB SNR and 22.8% at 10 dB SNR.

Results at scaling factor of 0 dB (Fig. 5): In this case, the ratio between LOS and NLOS components remains the same across both sub-6 GHz and mmWave frequency bands. The SE gain at a K-factor of 20 dB is 87.6% at 0 dB SNR and 22.6% at 10 dB SNR.

Results at scaling factor of 10 dB: Here, the sub-6 GHz K-factor is 10 dB lower than the mmWave K-factor. As the mmWave channel becomes more directive with a stronger LOS

TABLE II
NUMBER OF OPERATIONS

Method	Value
LS Estimation/Interpolation (mmWave)	$N^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})} \left(6 M_{\mathrm{Tx}}^{(\mathrm{m})} - 5\right)$
LS Estimation/Interpolation (sub-6 GHz)	$N^{(\mathrm{s})} M_{\mathrm{Rx}}^{(\mathrm{s})} \left(6 M_{\mathrm{Tx}}^{(\mathrm{s})} - 5\right)$
Joint AoD/AoA Estimation	$3 L_{\vartheta_{\mathrm{tr}}} L_{\varphi_{\mathrm{tr}}} \left(M_{\mathrm{Tx}}^{(\mathrm{s})} M_{\mathrm{Rx}}^{(\mathrm{s})}\right)^2$
Translating	$N^{(\mathrm{m})} M_{\mathrm{Tx}}^{(\mathrm{m})} \left(1 + M_{\mathrm{Rx}}^{(\mathrm{m})}\right) + 2 \left(N^{(\mathrm{m})}\right)^2$
Averaging	$M_{\mathrm{Rx}}^{(\mathrm{m})} M_{\mathrm{Tx}}^{(\mathrm{m})} \left(2 N^{(\mathrm{m})} + 1\right)$
MRC	$5 N^{(\mathrm{s})} M_{\mathrm{Rx}}^{(\mathrm{s})} M_{\mathrm{Tx}}^{(\mathrm{s})}$

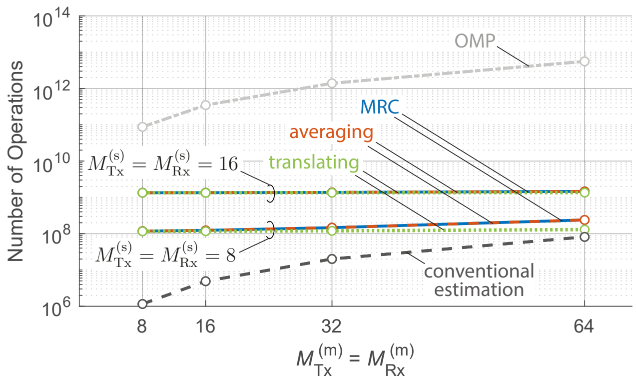

Fig. 9. Joint AoD/AoA estimation leads to a significant increase in the number of operations compared to conventional estimation. However, when comparing the proposed channel estimation methods against each other, the difference in the number of operations becomes significantly smaller.

component, the sub-6 GHz channel retains significant multipath components. This weakens the LOS estimation quality when leveraging sub-6 GHz information, negatively impacting the performance of the proposed method. The SE gain over the conventional method at a K-factor of 20 dB drops to 85.7% at 0 dB SNR and 20.1% at 10 dB SNR.

F. Computational Complexity

In practical applications, not only the performance of the proposed methods is important, but also their computational complexity. Therefore, we evaluate the computational complexity of the proposed methods by considering the number of mathematical operations required. The number of operations required is summarized in Table II and plotted as a function of the number of mmWave transmit/receive antennas in Fig. 9.

The conventional channel estimation method relies solely on LS estimation and interpolation within the mmWave band, as detailed in Section III-A. In contrast, the proposed channel estimation methods require LS estimation and interpolation at both, the sub-6 GHz and mmWave bands, as preliminary steps. Additionally, the proposed methods include a crucial step of joint AoD/AoA estimation, which significantly increases the number of operations compared to conventional estimation. Consequently, a 16×16 sub-6 GHz MIMO configuration poses significantly higher complexity compared to the 8×8 configuration. Moreover, the OMP-based channel estimation method

TABLE III
OVERHEAD ANALYSIS

Method		Overhead (Time-Symbols)
Analog Beamforming		$2 M_{\mathrm{Tx}}^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})}$
Hybrid Beamforming		$2 \frac{M_{\mathrm{Tx}}^{(\mathrm{m})}}{M_{\mathrm{RF,Tx}}^{(\mathrm{m})}} \frac{M_{\mathrm{Rx}}^{(\mathrm{m})}}{M_{\mathrm{RF,Rx}}^{(\mathrm{m})}}$
Digital Beamforming	Conventional	2
	OMP	2
	Proposed	4

incurs the highest number of operations due to the complexity and iterative nature of the algorithm (see [52]), which has to be applied to each of $N^{(\mathrm{m})}$ mmWave subcarriers.

However, when comparing the proposed channel estimation methods against each other, the difference in the number of operations becomes noticeably smaller. Both the MRC and averaging methods have a higher complexity in comparison to the translating method. This complexity arises from the fact that the translating method utilizes only the out-of-band aided mmWave LOS channel estimate, as detailed in Eq. (14). In contrast, the other two methods require the additional step of combining the out-of-band aided and in-band mmWave channel estimates, as specified in Eq. (37) and Eq. (38). Moreover, unlike the averaging method, the MRC method requires an additional step of estimating the K-factor and calculating the optimal combining factor, as detailed in Section V-C.

G. Overhead Analysis

In practical systems, the overhead of the proposed methods is an important consideration. Therefore, we evaluate the overhead by considering the number of mmWave training symbols (time-symbols) required for link establishment. The sub-6 GHz system does not introduce additional time overhead, as it operates simultaneously with the mmWave system. The training overhead is summarized in Table III.

Analog beamforming systems incur the highest overhead, as beam training is typically performed via an exhaustive search over $M_{\mathrm{Tx}}^{(\mathrm{m})} M_{\mathrm{Rx}}^{(\mathrm{m})}$ beam directions. Hybrid beamforming reduces this overhead by employing a limited number of transmit and receive RF chains ($M_{\mathrm{RF,Tx}}^{(\mathrm{m})}$ and $M_{\mathrm{RF,Rx}}^{(\mathrm{m})}$), allowing multiple beams to be tested simultaneously. Digital beamforming has the lowest training overhead due to its ability to perform parallel beamforming. This is achieved by multiplexing signals from different antennas in the frequency domain (see Fig. 3). Among digital beamforming architectures, conventional and OMP-based methods require only one time-symbol per link direction (transmitter-to-receiver or receiver-to-transmitter). Our proposed methods (translating, averaging, and MRC) introduce an additional time-symbol per link direction for LOS channel estimation (see Section II-B1). While this slightly increases the overhead compared to conventional and OMP methods, it remains negligible compared to analog and hybrid beamforming. Note that in the TDD system under consideration, link establishment at both ends (including precoder and combiner design) requires channel estimation at both the transmitter and receiver for optimal beamforming

Fig. 10. Measured outdoor environment. The transmitter and receiver are placed 27 m apart from each other.

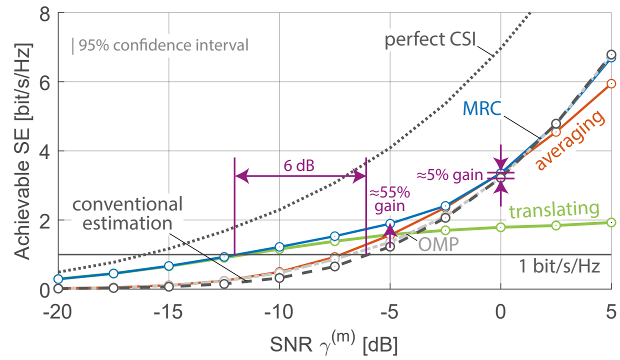

Fig. 11. The MRC method outperforms the conventional approach in achievable SE across the entire SNR range, with a 5% gain at an SNR of 0 dB and a 55% gain at an SNR of -5 dB. Our proposed MRC method effectively requires 6 dB less SNR compared to the conventional method to achieve a SE of 1 bit/s/Hz.

TABLE IV
CHANNEL-SOUNDING PARAMETERS

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	3 GHz	60 GHz
Wavelength λ	10 cm	0.5 cm
Bandwidth B	20.8 MHz	400 MHz
Number of Frequency Points N_meas	14	250
Virtual Antenna Configuration	8×8	8×8
Subcarrier Spacing $\triangle f_\mathrm{meas}$	800 kHz	800 kHz
Antenna Spacing	5 cm	0.25 cm

matrix selection. This results in a doubling of the training overhead (as reflected in Table III), with one training instance for the transmitter-to-receiver direction and another for the receiver-to-transmitter direction.

VII. Measurement-Based Comparison

In this section, we complement our simulations with a measurement-based evaluation of the proposed channel estimation methods. Section VII-A describes the multi-band MIMO channel-sounding campaign and the corresponding evaluation of the measurement results. Next, in Section VII-B, we incorporate these channel-sounding results into the simulations (as in Section VI) to validate the performance of the proposed channel estimation methods.

A. Measurement Campaign

The channel-sounding campaign is conducted in a LOS outdoor scenario, as shown in Fig. 10. This measurement environment corresponds to a device-to-device (D2D) communication scenario, where information is exchanged directly between nodes without involving a base station or access point. A dual-band channel sounder system is employed, utilizing a vector network analyzer (VNA) (KT-N5222A-200/WXP) with an 8×8 virtual MIMO antenna array configuration. The sounder operates within both sub-6 GHz (2−6 GHz) and mmWave (59−63 GHz) frequency bands, with a detailed description provided in [20], [21]. The transmitter and receiver are positioned 27 m apart, each using omni-directional biconical antennas. To ensure the comparability of the measurement results, identical locations (with unchanged surroundings) are measured for both frequency bands. The antennas at both the transmitter and receiver are moved to different positions along the horizontal plane by a linear positioner, creating a virtual ULA. Additionally, the transmit and receive antennas are maintained at the same height, ensuring that elevation differences have a negligible impact. The parameters utilized in this channel-sounding campaign are summarized in Table IV. Finally, we extract the channel transfer function (CTF) from the data captured by the VNA after each measurement run. As the result of the campaign, we obtain the measured CTF $\mathbf{H}^{(\mathrm{b})}_{\mathrm{meas}}[m] \in \mathbb{C}^{\mathrm{M}^{(\mathrm{b})}_{\mathrm{Rx}} \times \mathrm{M}^{(\mathrm{b})}_{\mathrm{Tx}}}$, where $m \in \{1, \ldots, N^{(\mathrm{b})}_{\mathrm{meas}}\}$.

The subcarrier spacing used in the channel-sounding campaign is 800 kHz, which is significantly larger than the subcarrier spacing employed in Section VI. To match the subcarrier spacing needed for our analysis, we apply sinc interpolation in the frequency domain to the measured CTF, given by

$$\mathbf{H}^{(\mathrm{b})}[n] = \sum_{m=1}^{N^{(\mathrm{b})}_{\mathrm{meas}}} \mathbf{H}^{(\mathrm{b})}_{\mathrm{meas}}[m] \,\mathrm{sinc}\left(\frac{f^{(\mathrm{b})}[n] - f^{(\mathrm{b})}_{\mathrm{meas}}[m]}{\triangle f^{(\mathrm{b})}}\right), \tag{57}$$

where $f^{(\mathrm{b})}[n] = (n-1)\triangle f^{(\mathrm{b})}$, and similarly defined, the measured grid $f^{(\mathrm{b})}_{\mathrm{meas}}[m] = (m-1)\triangle f^{(\mathrm{b})}_{\mathrm{meas}}$. This allows us to reduce the subcarrier spacing to $\triangle f^{(\mathrm{s})} = 60$ kHz for the sub-6 GHz system and $\triangle f^{(\mathrm{m})} = 120$ kHz for the mmWave system, thereby producing CTFs comparable to those in Section VI.

B. Performance as a Function of SNR

Next, we utilize the measured and interpolated CTFs $\mathbf{H}^{(\mathrm{s})}[n]$ and $\mathbf{H}^{(\mathrm{m})}[n]$ to evaluate the performance of the proposed channel estimation methods as a function of SNR, following the approach in Section VI-C. The evaluation is based on a single realization of the measured MIMO channel. The simulation results for the 8×8 MIMO configuration in terms of achievable SE as a function of the SNR are shown in Fig. 11. The measured SNR is 40 dB for the sub-6 GHz band and 5 dB for the mmWave band. To achieve varying SNR levels, complex Gaussian noise with adjustable power is added to the received signal. Furthermore, the estimated RMSE for the LOS AoA is approximately $2.31°$, while the estimated RMSE for the LOS AoD is approximately $1.53°$.

The translating method outperforms the conventional and OMP-based methods within the SNR range from -20 to -5 dB. Similarly, the averaging method performs better than the conventional and OMP-based methods over almost the entire SNR range (-20 to 0 dB). Finally, the MRC method consistently outperforms the conventional and OMP-based methods in terms of achievable SE across the entire SNR range, showing a 5% gain at an SNR of 0 dB and a 55% gain at an SNR of -5 dB. Notable improvement is observed at low SNRs, while at higher SNRs, the MRC method's performance gradually converges with that of the conventional and OMP-based ones. For example, to support a SE of 1 bit/s/Hz, the conventional method required an SNR of -6 dB. In contrast, with the proposed MRC method, the same SE is achieved at an SNR of -12 dB. This means our proposed MRC method effectively requires 6 dB less SNR compared to the conventional method for the same performance.

VIII. Conclusion

In this paper, we propose a novel approach for estimating CSI in mmWave MIMO systems leveraging out-of-band CSI from the sub-6 GHz band. The proposed approach relies on AoA and AoD estimation and exploiting the relationship between LOS channel components across different frequency bands. Based on this approach, we develop three ways of combining conventional in-band mmWave CSI and out-of-band reconstructed CSI. Numerical simulations confirm the effectiveness of the proposed channel estimation methods in comparison to conventional and sparse OMP-based methods that utilize only in-band mmWave information. The SNR and K-factor affect the performance of the proposed channel estimation methods to a great extent. The proposed methods lead to a significant increase in achievable SE in the low-SNR regime, while the gains are negligible in the high-SNR regime. The translating method performs well at high K-factors, while the averaging method performs well at moderate K-factor. The MRC method is better (high K-factor) or at least as good (low K-factor) as the conventional channel estimation method using only the mmWave band. Furthermore, increasing the number of mmWave antennas relative to sub-6 GHz antennas results in a reduction in achievable SE of the proposed methods. Consequently, for large mmWave MIMO configurations, the sparse channel estimation method based on the OMP algorithm outperforms our proposed methods. However, it is worth noting that while the proposed channel estimation methods increase computational complexity compared to conventional estimation, their complexity remains lower than that of the method using the OMP algorithm. Although the proposed methods pose slightly increased overhead compared to the conventional and OMP-based method, this overhead remains negligible compared to analog and hybrid beamforming, as our approach utilizes digital beamforming.

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